PhAIL: A Real-Robot VLA Benchmark and Distributional Methodology

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Abstract

How do you tell whether one robot policy is better than another in the real world? Most published evaluations of vision-language-action (VLA) policies answer this with binary success rate at a fixed timeout, no confidence intervals, no paired test, $N \leq 25$ per condition – under-budgeted by roughly two orders of magnitude for the ranking claims it publishes. We respond with a distributional methodology for VLA evaluation, and release PhAIL – a real-world inference benchmark that instantiates it. The time-to-success CDF is the evaluation primitive, absorbing hard failures (drops outside the workspace, safety stops) as $T=\infty$ ghost events. Reporting and ranking are separated: a scalar with bootstrap intervals (Human-Relative Throughput, HRT) for reporting; paired Kolmogorov–Smirnov on the CDF for ranking. On a Franka FR3 across four objects with $\sim$950 episodes and $\sim$35 paired rollouts per (model, object) cell, we evaluate four frontier VLAs (OpenPI $\pi_{0.5}$, GR00T N1.6, ACT, SmolVLA): the best frontier model is roughly $7\times$ slower than a human teleoperator, the top two are statistically indistinguishable, and SmolVLA is clearly worst. Two structural findings: different reasonable scalars rank the same models in different orders, and a distributional test resolves the closest ranking with $\sim 30\times$ fewer trials than binary success rate. Benchmark, dataset, and reference implementation: URL withheld for double-blind review.

1 Introduction

How do you tell whether one robot policy is better than another in the real world? This question is the gate to every deployment claim in robotics, and the answer the field has converged on is structurally insufficient for the claims it makes. Most published evaluations of vision-language-action (VLA) policies report binary success rate at a fixed timeout, with no confidence intervals and no paired test, at $N \leq 25$ rollouts per condition. As VLA development accelerates and frontier models begin to compete head-to-head on real hardware, getting evaluation right is what tells the field whether progress is real or the rankings are noise.

The standard fix the field has reached for – adding a throughput metric (UPH, cycle time, completion fraction) alongside binary success – works in part but surfaces a deeper problem: when the two scalars rank policies in different orders, the choice between them is task-dependent and rarely disclosed in print. Recent methodology critiques argue that blinded same-session A/B with hundreds of rollouts is feasible [22, 23], and the closest ML-side precedent for distributional comparison [24] extends the conversation toward distributions. None of these recommendations ships a primitive plus a reference implementation that lets the field apply them on a shared station.

We respond by adopting the time-to-success CDF as the evaluation primitive. Let T be the wall-clock time the policy needs to complete one operation; hard failures (items lost outside the workspace,

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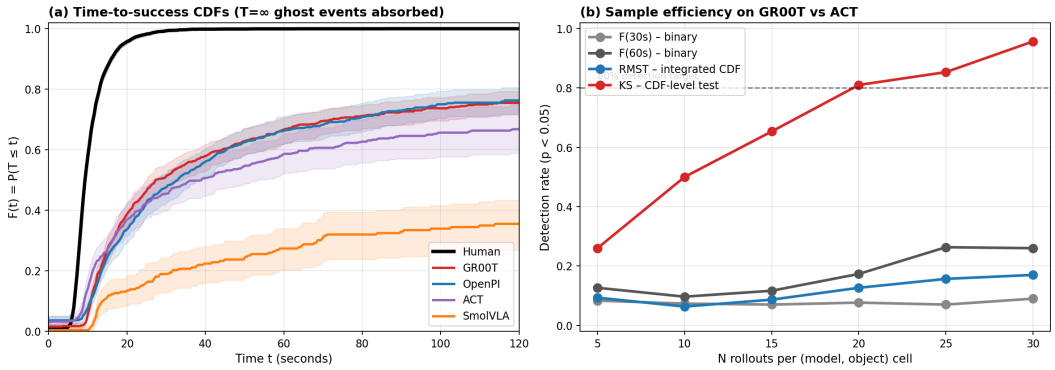


Figure 1: **(a)** Time-to-success CDFs are richer than any scalar: reliability and throughput on a single axis. The four frontier VLAs all sit far below the human reference – the best is $\sim 7\times$ slower. Hard failures (drops outside the workspace, safety stops) become the asymptote below $F = 1$ instead of disappearing into a censored tail. **(b)** Choosing the right *test*, not just running more trials, is what resolves close rankings: paired Kolmogorov–Smirnov on the macro CDF reaches 80% detection on the borderline pair GR00T vs. ACT at N=20 paired rollouts per (model, object) cell, while binary success-rate-at-threshold ($F(30s)$, $F(60s)$) and integrated RMST stay near the noise floor across the entire $N \leq 30$ range. The dominant practice ($F(\tau)$ at modal $N \leq 25$) is structurally too coarse to back the rankings it publishes.

items dropped on the table at episode end, safety stops) are absorbed as $T = \infty$ ghost events rather than censored tails. The resulting CDF $F(t) = P(T \leq t)$ jointly carries reliability (its asymptote below 1 equals the hard-failure rate) and throughput (the median of T, and UPH $\propto 1/E[T \mid T < \infty]$). Standard scalars – success rate at τ, UPH, MTBF/A, completion fraction, RMST – are projections of F, and the CDF is strictly richer than any of them: when two CDFs cross, no single scalar suffices and reasonable scalars give opposite signs on the same data. We separate two methodological jobs the field blurs together: *reporting* (a scalar with bootstrap CIs – Human-Relative Throughput, HRT, anchored to a same-fixture human reference) and *ranking* (a paired Kolmogorov–Smirnov test on the CDF). PhAIL is the open real-world inference benchmark that instantiates this methodology on a Franka FR3 across four objects.

Empirically, on four frontier VLAs – OpenPI $\pi_{0.5}$ [28], GR00T N1.6 [29], ACT [30], SmolVLA [31] – the best is roughly $7\times$ slower than the human reference, the top two are statistically indistinguishable on every metric tested, and SmolVLA is clearly worst. Two structural findings result. First, principled aggregation rules over the same CDF (RMST vs. AUC-vs-human) yield opposite top-1 rankings – no single scalar uniquely encodes operational priority. Second, the choice of *test*, not just the choice of N, determines what is resolvable: paired KS detects the GR00T-vs-ACT ranking at N=20 paired rollouts per cell, while binary success rate at a fixed threshold needs roughly $30\times$ more rollouts (600–1500 paired by McNemar) to reach the same 80% detection. The field's modal $N \leq 25$ per condition is two orders of magnitude under-budgeted for the ranking claims it publishes.

Contributions.

1. **Time-to-success CDF as the evaluation primitive**, with hard failures absorbed as $T = \infty$ ghost events (Figure 1a). Standard scalars are projections; we propose macro-averaged Human-Relative Throughput (HRT) as the dimensionless headline scalar grounded in a same-fixture human reference.
2. **Same data, opposite rankings – no single scalar suffices.** Principled aggregations of the same CDF disagree on top-1, so the choice of headline scalar is a disclosed methodological commitment.
3. **Distributional tests are roughly $30\times$ more sample-efficient than binary success rate** (Figure 1b). The choice of test, not just the choice of N, is the bigger lever; reporting (HRT) and ranking (paired KS on the macro CDF) should be separate.

4. **PhAIL: open real-world benchmark and reference implementation.** Four frontier VLAs, four trained objects, ∼950 episodes including 386 human reference rollouts, full per-rollout artifacts, open-source evaluation framework end-to-end.

Scope. The methodological contribution is the comparison primitive, the ranking procedure, the power budget, and the open framework that makes new claims auditable. The methodology is task-agnostic – it applies to any operation with an unambiguous success event and a same-fixture human reference. PhAIL's empirical validation is bin-to-bin pick-and-place across four objects (§4); other natural fits include insertion, packing, navigation, and articulated manipulation. The per-model rankings are illustrative of what the methodology can resolve at our depth, not architectural pronouncements; we used each repository's default fine-tuning recipe (§4.2). PhAIL is open to submissions.

2 Related Work

Existing real-robot evaluation protocols. RoboArena [12] and RoboChallenge [11] use physical robots but compare via pairwise Elo or scalar success on standardized tasks; neither reports confidence intervals or paired tests on the rankings. Competition-style evaluations (RGMC, OCR-TOC [18], NIST Assembly Task Boards [19], the Real Robot Challenge [20], surveyed in [17]) standardize task scoring without statistical machinery; closest in spirit is the Digital Robot Judge [21], which builds task-centric performance databases via electronic task boards but reports raw telemetry only. The most rigorous existing real-robot evaluation is the LBM examination [23], which runs blinded same-session A/B at N=50–200 per condition with Bayesian posterior violins on success rate, paired Barnard's exact tests for binary outcomes, Welch's t-tests for a scalar task-completion score, and Bonferroni-corrected significance grouping – a strong precedent for what rigorous real-robot evaluation looks like. PhAIL's contribution is orthogonal to LBM's: a richer primitive (time-to-success CDF with T=∞ ghost events, jointly carrying reliability and throughput) and a same-fixture human-reference anchor that turns embodiment-specific timings into a dimensionless ratio. Simulated benchmarks (RoboCasa, CALVIN, ManiSkill, SIMPLER [13, 14, 15, 16]) iterate fast but miss real-world perception, latency, and safety; they are complementary, not substitutes.

Methodology critiques. Kress-Gazit et al. [22] argue that current real-robot evaluation is statistically underpowered and recommend blinded same-session A/B at hundreds of rollouts; STEP [26] concurs ("20–30 real-world trials are insufficient for statistically significant conclusions"). On the ML side, Agarwal et al. [24] are the closest precedent for distributional comparison in evaluation. We extend their distributional approach to time-to-event data with T=∞ ghost events and ship a reference implementation of Kress-Gazit's recommendations on a shared station. A 13-paper survey of recent VLA evaluation practice (Appendix A) confirms the same pattern: modal per-condition N is 10–20, and none of the 13 standard-practice papers report confidence intervals or paired tests; LBM is the single recent counter-example.

3 Time-to-Success CDF as Evaluation Primitive

The methodology has three components: the random variable and its CDF (§3.1), the reporting layer (scalars and visualizations derived from the CDF, §3.2), and the ranking layer (paired tests on the CDF, §3.3). The strict separation between reporting and ranking is the methodological core. We use concrete language drawn from our running bin-to-bin pick-and-place setup (formal description in §4) – successful placements, items dropped outside the workspace – but the framework applies unchanged to any operation with an unambiguous success event, a measurable time-to-success, an identifiable notion of *hard failure* (states from which the policy cannot recover), and a same-fixture human-reference baseline.

3.1 The Random Variable T

For a single rollout episode with multiple operations to perform, we extract a sequence of times-to-success T_i and event indicators $E_i \in \{0, 1\}$. A successful placement contributes T_i equal to the elapsed time since the previous successful placement (or since episode start, for the first), with $E_i = 1$. If the episode ends with operations remaining (operator stopped, timed out,

model stalled), the unfinished attempt contributes one right-censored pair $(T_{\text{tail}}, 0)$ where $T_{\text{tail}} =$ duration $-$ last placement time; $E = 0$ means we know $T > T_{\text{tail}}$ but do not observe T.

Hard failures as $T = \infty$ ghost events. Some operations end in states the policy cannot recover from. The clearest case is an item dropped outside the workspace, where the item is physically gone from the scene. Items dropped within the workspace but outside both totes (e.g., on the table) are a softer category: in principle the policy can pick them back up, and we observed at least one rollout where this happened. We count those as failures only if they are still uncollected at episode end: as many failures as there are dropped-and-not-recovered items at the time the rollout terminates. A safety-stop termination is a third class: the rollout ends because the operator stopped it, and the operation that triggered the stop contributes one ghost event. We absorb all of these as $(T = \infty, E = 1)$ ghost events: the operation *did* terminate (so $E = 1$, not censored), but at $T = \infty$ rather than at any finite time. The CDF carries this as an asymptote below $F = 1$; the asymptote's height equals the hard-failure rate. Stalls and timeouts are explicitly *not* ghost events: those are slow-but-recoverable runs that belong in the censored tail, where they correctly indicate "the policy was working when we stopped watching it."

Estimation and censoring. The CDF $F(t) = P(T \leq t)$ is estimated by the Kaplan-Meier product-limit estimator [1] on the (T, E) collection. Confidence intervals on derived scalars (RMST, HRT, $F(\tau)$ at fixed thresholds) are reported via 95% episode-clustered bootstrap [4]. Episodes with zero successes contribute a single right-censored observation of length τ_{episode}, the per-rollout time budget after which an unfinished operation is censored (set by the protocol; §4.2); we never drop them. The $T = \infty$ atom is handled directly by the estimator (a discrete jump at the right edge); for the integrated-metric calculation in §3.2 we set $T_{\text{ghost}} \to \tau$, the integration cap, so a hard failure inflates RMST by exactly τ minus the time the model would have spent on that operation if successful.

3.2 Reporting: Scalars and Visualizations from the CDF

The CDF is the lossless representation. Every standard scalar in published VLA evaluations is a projection of it (Table 1); reporting any single one without the CDF discards information.

Scalar	Functional of $F(t)$
Success rate at τ	$F(\tau)$
Median time-to-success	$\inf\{t : F(t) \geq 0.5\}$
RMST(τ) (restricted-mean survival time)	$\int_0^\tau \big(1 - F(t)\big)\, dt$
UPH (units per hour)	$3600/E[T \mid T < \tau_{\text{episode}}] \propto 1/\text{RMST}(\tau_{\text{episode}})$
Completion fraction	$F(\tau_{\text{episode}})$
MTBF/A (mean time between failures/assists)	$\text{RMST}(\tau_{\text{episode}})/\big(1 - F(\tau_{\text{episode}})\big)$

Table 1: Standard scalar metrics expressed as functionals of the time-to-success CDF $F(t)$. The per-rollout time budget τ_{episode} is the protocol's timeout (§4.2); $1 - F(\tau_{\text{episode}})$ is the hard-failure rate (operations that did not complete within budget, including $T = \infty$ ghost events).

Headline scalar: Human-Relative Throughput (HRT). For each (model, object) cell we compute

$$\text{HRT}(m, o) = \frac{\text{RMST}_{\text{ref},\, o}(\tau)}{\text{RMST}_{m,o}(\tau)}, \qquad \tau = 240\,\text{s},$$

the cell-wise ratio of human-reference to model RMST, reported as a percentage, and macro-aggregated across objects with equal weights. Three properties make this the right scalar to report. (i) It inherits the CDF's joint speed-and-completion property via $T = \infty$ inflating $\text{RMST}_{\text{model}}$. (ii) It is grounded in operator practice (UPH-equivalent against a same-fixture human reference, so embodiment confounds cancel). (iii) Cell-wise normalization absorbs object-difficulty differences – a hard object slows the human and the policy proportionally – yielding a dimensionless quantity comparable across heterogeneous tasks. Cross-deployment comparison (different operators, different rooms, different reference pacing) reduces to comparing ratios, not absolute throughputs.

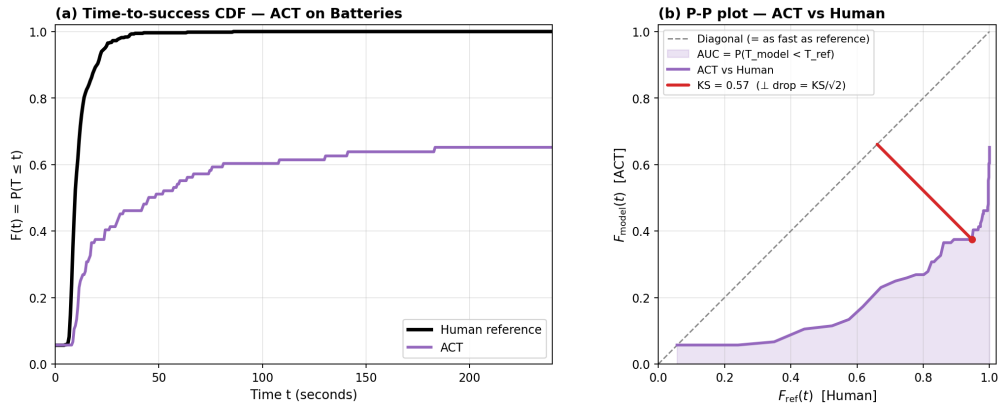


Figure 2: One (model, object) cell (ACT on Batteries) illustrating the reporting layer: time-to-success CDFs (left) and the corresponding P-P plot (right). **Left:** model and human-reference CDFs on the same axis; the asymptote below $F = 1$ on the model curve is the hard-failure rate. **Right:** $\big(F_{\text{ref}}(t),\, F_{\text{model}}(t)\big)$ traced parametrically as t sweeps. Diagonal = as fast as the reference; below-diagonal = slower. The shaded area under the curve is the AUC $P(T_{\text{model}} < T_{\text{ref}})$; the perpendicular distance from the curve to the diagonal at the point of maximum separation (red segment) equals $\text{KS}/\sqrt{2}$, where KS is the Kolmogorov-Smirnov statistic.

Visualization: P-P plots. For a (model, object) cell, the P-P plot traces $\big(F_{\text{ref}}(t),\, F_{\text{model}}(t)\big)$ parametrically as t sweeps from 0 to τ. The diagonal is "as fast as the reference"; below-diagonal is slower; above-diagonal is faster. Two derived properties give the P-P curve its analytic value. The area under the curve equals $P(T_{\text{model}} < T_{\text{ref}})$, the pairwise stochastic-dominance probability (AUC). The maximum vertical distance from the curve to the diagonal equals the Kolmogorov-Smirnov distance between the two CDFs (§3.3); the P-P plot is therefore the natural visualization for both. Figure 2 shows one (model, object) cell's CDF and the corresponding P-P plot side by side. P-P plots make distributional shape legible (saturation, early-success peaks, tail behavior); we use them descriptively in §5.3.

Confidence intervals. All reporting scalars and CDF curves carry 95% paired episode-clustered bootstrap intervals [4]. We resample whole episodes (not operations within episodes) because operations within an episode are correlated by the rollout's shared visual conditions and policy state; per-operation bootstrap produces fake-tight intervals.

3.3 Ranking: Paired Tests on the CDF

Reporting a scalar with a CI is one job; deciding whether two policies' underlying distributions differ – and if so, in what direction – is a different job. We treat it as such. By *ranking* we mean the joint claim of significance plus direction, with three possible outcomes: **(i)** A **is better than** B (CDFs separate, consistently signed – one dominates the other across t); **(ii)** A **and** B **differ but neither is uniformly better** (CDFs cross – reasonable scalars give opposite signs because they weight different regions of the curve, so disagreement among scalars is the CDF's signal that the policies have different distributional shapes, not a scalar bug); **(iii)** A **and** B **are indistinguishable** at the available N.

Our primary test for the significance gate (regimes (i)+(ii) vs (iii)) is the two-sample Kolmogorov–Smirnov statistic [3]

$$D = \sup_t \big|F_a(t) - F_b(t)\big|,$$

computed per-object on the macro-aggregated CDF, with p-values from a paired episode-clustered bootstrap [4] pooling the two policies' subsamples per object under $H_0 : F_a = F_b$. Geometrically, D is the maximum vertical distance from the P-P plot of F_a versus F_b to the diagonal (Figure 2). As a sanity-check we report stratified pooled-across-objects logrank [2] with Bonferroni correction over the model pairs. Direction (outcome (i) vs (ii)) comes from a paired scalar – in our case the

macro RMST difference, equivalent to signed Wasserstein-1 in 1D – with outcome (ii) surfaced when reasonable scalars disagree (§5.3).

The two-jobs framing – scalar for reporting, KS for the significance gate – is non-trivial. §5.2 demonstrates empirically that RMST-as-scalar is not measurably more sample-efficient than $F(60\,\mathrm{s})$ for distinguishing distributions, while KS-as-test materially is. The information lives in *where the curves separate*; an integrator smears that information, a sup-norm test catches it.

4 PhAIL: Design, Protocol, and Power Budget

4.1 Platform and Task

The evaluation station uses a DROID-style configuration [10]: a Franka Research 3 with a Robotiq 2F-85 parallel-jaw gripper, plus an over-the-shoulder external RGB camera and a wrist-mounted RGB camera. The first task is bin-to-bin order picking: items move one at a time from an inbound (transparent) tote to an outbound (grey) tote across four object types spanning a useful slice of physical regimes – wooden spoons (rigid, elongated, multi-grasp), towels (deformable), scissors (articulated, metallic), and batteries (small, rigid). Each rollout episode contains many independent placement operations, which we exploit for sample size; the framework applies just as well to single-operation episodes. Bin-to-bin is deliberately not the most complex task we could put VLAs through; it is the simplest task with unambiguous success/intervention semantics that still carries clear commercial signal (millions of picks per day across warehousing and fulfilment). Subsequent PhAIL releases will add new tasks (insertion, small-part assembly); the methodology is unchanged across additions. The same hardware operates the human reference and every evaluated VLA via an open-source evaluation framework [9] behind a common station interface, so only weights and architecture-specific observation-to-action code differ across submissions. Every evaluated rollout is published with synchronized video, telemetry, automated event labels, and crowd-sourced annotations, browsable in a run-explorer interface (Figure 6, Appendix C; annotation pipeline in Appendix G). Benchmark, dataset, and per-rollout artifacts: URL withheld for double-blind review.

4.2 Models, Fine-Tuning, and Protocol

We evaluate four publicly available VLAs: **OpenPI** $\pi_{0.5}$ [28] (3B-param, FAST action tokenization on flow-matching), **GR00T N1.6** [29] (3B-param, Cosmos-Reason VLM + 32-layer diffusion transformer action head), **ACT** [30] (Action Chunking Transformer, CVAE), and **SmolVLA** [31] (450M-param, HuggingFace/LeRobot). None has zero-shot capability on this task, so fine-tuning is a prerequisite. All four were fine-tuned on the same 352-episode bin-to-bin demonstration set ($\sim$13 hours; Appendix D) using each repository's default recipe – no per-model hyperparameter search. **This is a deliberate methodological constraint, not best practice for production deployment;** per-model recipe optimization could shift the rankings of §5.

Each rollout has a 30 s/item time cap ($\sim$10$\times$ the $\sim$2.7 s/item human pace). Evaluation is blind: the scheduler randomly selects which checkpoint runs next, the operator does not know which is running, and the operator's only intervention is triggering a safety stop. Post-rollout, the operator records target/source counts, items lost outside both bins, and outcome label $\in$ {Success, Stalled, Safety, Ran_out_of_time}; every successful placement is annotated post-hoc from the rollout video (Appendix G).

4.3 Cohort and Power Budget

After object/model filtering, $\sim$950 episodes enter the analyses; 386 are human reference rollouts. Annotations come from manual review of rollout video augmented by automated success detection (Appendix G); both label streams are released alongside the data.

How many rollouts does a ranking claim actually need? Standard power calculations (Appendix F) put the field-modal N=10–20 (Appendix A) orders of magnitude under-budget: a $\pm$5 pp Wilson CI on a single arm needs $N\approx$380, and detecting a 5 pp paired difference between two policies (McNemar, 80% power, α=0.05) needs 600–1500 paired rollouts. Our own $N \approx 35$ per cell is 2–3$\times$ the field median but still well below either threshold; §5.2 confirms this empirically with detection-rate-vs-N curves on our data.

5 Results

All numbers below use $\tau = 240\,\text{s}$ for RMST, $n_{\text{boot}} = 1000$ for episode-clustered bootstrap CIs, seed 0. Per-(model, object) cell sizes are 30–45 episodes for OpenPI, GR00T, and ACT; 24–29 for SmolVLA.

5.1 Headline Ranking

Model	RMST (s)	HRT (%)	Intervention rate	Episodes
Human reference (teleop)	10.5	100.0	—	392
NVIDIA GR00T N1.6	78.3	**13.5**	2.3%	171
Physical Intelligence Open $\pi_{0.5}$	81.1	13.0	4.5%	177
Action Chunking Transformer	98.2	10.7	0.7%	148
Hugging Face SmolVLA	166.9	6.3	16.3%	123

Table 2: Headline ranking ($\tau = 240\,\text{s}$, $N \approx 35$/cell). HRT is the cell-wise reference-to-model RMST ratio, macro-averaged across four trained objects. Intervention rate is (safety stops + stalls) / total episodes (lower bound; drops outside the workspace not included).

GR00T and OpenPI are within 0.5 percentage points of HRT, well inside any reasonable noise floor at this N. No inference model exceeds 19% HRT on any single object (per-cell RMST grid, Figure 4). Even the best frontier VLA is roughly 7× slower than the human on the same fixture.

Pair	p (raw)	p (Bonferroni)	Significance
OpenPI vs. SmolVLA	$<10^{-15}$	$<10^{-14}$	***
GR00T vs. SmolVLA	$<10^{-15}$	$<10^{-14}$	***
ACT vs. SmolVLA	3.3×10^{-16}	2.0×10^{-15}	***
GR00T vs. ACT	2.3×10^{-3}	1.4×10^{-2}	*
OpenPI vs. ACT	1.1×10^{-2}	6.7×10^{-2}	borderline
OpenPI vs. GR00T	4.5×10^{-1}	1.00	n.s.

Table 3: Pooled-across-objects logrank, Bonferroni-corrected over the 6 model-pairs.

The top two inference models cluster as statistically indistinguishable; ACT is separable from GR00T and borderline from OpenPI; SmolVLA is distinguishably worst. **Even at $N \approx 35$/cell – already 2–3× the field-modal N – the closest pair (OpenPI vs. GR00T) remains unresolved by pooled logrank.**

What the field's standard methodology would have said. At the modal τ=30 s, the binary leader is GR00T (0.51) with OpenPI (0.46) and ACT (0.44) close behind. Under AUC-vs-human the leader is *ACT* (§5.3); under RMST it is GR00T with OpenPI a hair behind, ACT a clear third. Same data, three principled scalar choices, three different top-1s – and Table 4 shows no binary-threshold metric can distinguish the close pairs on the underlying CDF at $N \leq 30$. A field-standard τ=30-second leaderboard at N=15 would publish whichever top-1 the random subsample favored, with no power to defend the choice.

5.2 Sample Efficiency: The Right Test, Not the Right Scalar

The headline above leans on RMST and pooled logrank. We now ask the operational question: at fixed N, which metric most reliably detects a ranking difference? Three contestants are tested side-by-side: binary $F(30\,\text{s})$ / $F(60\,\text{s})$ (the field's current practice), RMST as a scalar (a CDF-aware reporting scalar applied to ranking), and paired KS on the CDF (the distributional test we propose for ranking). For each pair of inference models and each $N \in \{5, 10, 15, 20, 25, 30\}$ per (model, object) cell, we ran 300 outer subsampling trials. Each trial subsamples N episodes per cell with replacement, then runs a 200-rep inner episode-clustered bootstrap to compute a two-sided p-value for the difference of each candidate metric (macro-averaged across objects, RMST and KS computed

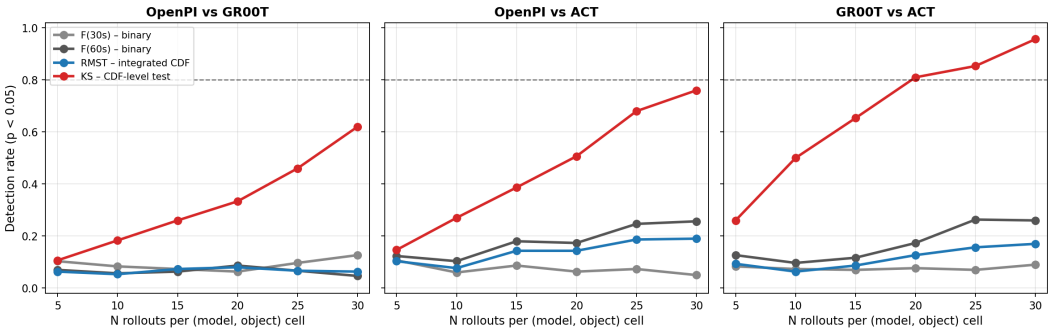


Figure 3: Detection rate vs. subsample size N per (model, object) cell, on the three close pairs that carry efficiency information (left: OpenPI vs. GR00T, the closest pair; centre and right: the two next-closest). Dashed line at 0.8 marks the conventional power target. KS (red) climbs steeply on every close pair while $F(30\,\mathrm{s})$, $F(60\,\mathrm{s})$, and RMST stay near the floor; KS reaches the 80% target within budget only on GR00T vs. ACT (right, N=20), climbing to 0.75 on OpenPI vs. ACT and 0.61 on OpenPI vs. GR00T at N=30 while the others stay below 0.30 across the entire range. The three any-vs-SmolVLA pairs saturate at $N = 5$ for every metric and are deferred to Appendix B. **The choice of test, not just the choice of N, determines what is resolvable.**

at $\tau = 120\,\mathrm{s}$ for the efficiency experiment). Detection rate is the fraction of trials with $p < 0.05$. Figure 1b shows the result for the borderline pair GR00T vs. ACT; Figure 3 shows the three pairs that carry efficiency information.

Pair	$F(30\,\mathrm{s})$	$F(60\,\mathrm{s})$	RMST	KS
OpenPI vs. GR00T	—	—	—	—
OpenPI vs. ACT	—	—	—	—
OpenPI vs. SmolVLA	5	5	5	10
GR00T vs. ACT	—	—	—	**20**
GR00T vs. SmolVLA	5	5	5	10
ACT vs. SmolVLA	5	5	5	10

Table 4: Smallest tested N at which detection rate ≥ 0.8 per (pair, metric). "—" means the detection rate did not reach 0.8 within the tested range $N \leq 30$. KS is the only metric that crosses the threshold within budget on a close pair (GR00T vs. ACT, N=20); on the other two close pairs KS reaches detection rates of 0.75 (OpenPI vs. ACT) and 0.61 (OpenPI vs. GR00T) at N=30, while every binary-threshold metric stays below 0.30.

The table reads in three regimes. **Three saturated pairs** (any inference model vs. SmolVLA): all metrics work; effects are large; binary baselines are slightly faster than KS at saturation because the pooled-resample H_0 calibration is conservative when effects are huge. **Three close pairs** (the three top-3 model pairs) – the bucket where sample efficiency actually matters. Only GR00T vs. ACT reaches the 80% target within budget, on KS at N=20. KS detection rate at $N = 30$ on the other two pairs is 0.75 (OpenPI vs. ACT) and 0.61 (OpenPI vs. GR00T), while $F(30\,\mathrm{s})$, $F(60\,\mathrm{s})$, and RMST stay below 0.30 on every close pair. KS dominates the binary baselines by a factor of 3–6 in detection rate at $N \leq 30$ across the three close pairs; resolving them to 80% empirically would require the 600–1500 paired rollouts the analytical power calculation predicts (§4.3).

Detection rate at $N = 30$ averaged across all six pairs: $F(30\,\mathrm{s}) = 0.54$, $F(60\,\mathrm{s}) = 0.60$, RMST $= 0.57$, **KS** $= 0.89$. The $+29$ pp gain of KS over $F(60\,\mathrm{s})$ is concentrated entirely on the three close pairs.

Two takeaways. First, **the distributional test (KS) is materially more sample-efficient than any single CDF-derived scalar (RMST or binary thresholds) for ranking.** RMST-as-scalar is essentially tied with $F(60\,\mathrm{s})$; the integration over $[0, \tau]$ does not buy ranking power because the information lives in *where the curves separate* and an integrator smears it. This does not invalidate

RMST as our recommended *reporting* scalar (HRT is built on it) – it just shouldn't carry the burden of *ranking* too: precisely the two-jobs framing of §3. Second, **empirical and analytical evidence triangulate.** The closest pair is unresolvable at $N = 30$ by every metric, matching the analytical 600–1500-paired-rollout requirement.

5.3 Aggregation-Rule Disagreement

Both **integrated RMST** and **AUC vs. human reference** are principled scalars derived from the *same* CDF; the choice of which to extract is itself a methodological commitment. Macro-AUC across objects: ACT 0.138 [0.111, 0.167] $>$ OpenPI 0.101 [0.084, 0.118] $\approx$ GR00T 0.096 [0.080, 0.112] $>$ SmolVLA 0.027 [0.017, 0.038]. ACT is now *highest* among inference models; under integrated RMST/HRT it sits third behind GR00T and OpenPI. Same data, two principled aggregations, opposite top-1s.

This is outcome (ii) of §3.3: ACT and OpenPI/GR00T are different (KS rejects) but neither is uniformly better – their CDFs cross, with ACT winning the early-time region (high AUC) and OpenPI/GR00T winning the integrated region (low RMST). The disagreement is sharpened by the human reference being $\sim 7\times$ faster than any evaluated VLA, collapsing the AUC integral onto the short-time region where ACT's early-success peak wins more first-success races. **When CDFs cross, no single scalar can summarize the comparison without weighting one region over another.**

We recommend **HRT** as the default headline scalar – it inherits the CDF's joint reliability-throughput property and matches operator-facing UPH practice; on a benchmark where reference and policies operated at comparable speeds, AUC-vs-human would track RMST closely. Pairwise model-vs-model AUC (Figure 5, Appendix B) confirms the RMST-side conclusion that the top three are indistinguishable; per-(model, object) RMST (Figure 4, Appendix B) shows the GR00T-vs-ACT object crossing visible at our N but not clearing Bonferroni at $\alpha = 0.05/24$.

6 Discussion and Conclusion

We introduced PhAIL, an open real-robot benchmark and distributional evaluation methodology for vision-language-action policies. The methodology adopts the time-to-success CDF as primitive, separates *reporting* (Human-Relative Throughput) from *ranking* (paired Kolmogorov–Smirnov on the macro CDF), and anchors to a same-fixture human reference. Evaluating four frontier VLAs across four objects, two structural findings emerged: principled aggregation rules over the same CDF yield opposite top-1 rankings, and paired KS detects ranking differences roughly $30\times$ more sample-efficiently than binary success-rate baselines.

Limitations. Single embodiment (Franka FR3, fixed cameras), single primitive (bin-to-bin pick-and-place, four trained objects); the framework generalizes to any operation with a human reference and event-time outcome but the empirical validation does not. $N{\approx}35$ per cell is 2–3$\times$ the field median but still below what binary metrics need to rank close pairs (§5.2); per-model rankings are illustrative. The single protocol recommendation that does the most work is **blind, randomized rotation on the same fixture in the same session** – spatial-configuration shifts (Appendix E) move outcomes by margins comparable to gaps between adjacent models, contaminating the CDF if not controlled.

If the field adopts this methodology, three things change. **Scalar choice becomes a disclosed commitment** (HRT and AUC-vs-human encode different priorities). **Ranking claims become falsifiable** via paired KS: "we cannot rank these at this N" becomes a publishable finding. **The hidden cost of binary thresholds becomes visible**: on close pairs, no realistic N recovers what a CDF-level test resolves within budget. PhAIL (URL withheld for double-blind review) is open to submissions; insertion and small-part assembly are next.

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A Survey of Recent VLA Evaluation Practice

Table 5 surveys 13 recent real-robot VLA papers from 2023–2025; the LBM examination [23] is included below the rule as the single recent counter-example. Modal per-condition N is 10–20; none of the 13 standard-practice papers report confidence intervals or paired tests. LBM reports Bayesian posterior credible regions and paired Barnard's exact / Welch's t-tests with Bonferroni-corrected significance grouping at $N{=}50$ real / 200 simulation per condition.

Paper	Per-condition N	Stats reported
RT-1 [32]	per-task N not standardized; $\sim$3000 total	Point estimates only
RT-2 [33]	$\sim$10–15 per task; $\sim$6000 total	Point estimates only
RT-X / Open X-Embodiment [34]	10–20 modal; up to 100/skill	Point estimates
OpenVLA [35]	$\sim$10 real / 50 sim per task	No CIs
π_0 [36]	$\sim$10 trials/task; partial credit	No CIs
$\pi_{0.5}$ [28]	tens per task, variable	No CIs
$\pi_{0.6}$ [37]	tens per task, variable	No CIs
GR00T N1 [29]	5 objects $\times$ 3 trials $=$ 15/task real	Point estimates
ACT / ALOHA [30]	$\sim$25 trials/task	Point estimates
Mobile ALOHA [38]	20/task (5 for "Cook Shrimp")	Point estimates
DROID [10]	10 rollouts per (task, method)	"Standard error" mentioned, no CIs
SmolVLA [31]	tens per real task	Point estimates
ECoT [39]	$\sim$20/task ($\sim$300 total / policy)	Point estimates
LBM Examination [23]	50 real / 200 sim per condition	Bayesian posteriors + paired Barnard / Welch with Bonferroni

Table 5: Per-condition N and statistical reporting in recent real-robot VLA evaluation papers. "Per-condition N" is the per-condition sample size for the modal task – the count of trials per (model, task) cell. "Stats" summarizes confidence-interval and paired-test reporting in the original paper.

Concurrent methodology critiques. Kress-Gazit et al. [22] explicitly argue that current real-robot evaluations are statistically underpowered. STEP [26] concurs: "20–30 real-world trials are insufficient for statistically significant conclusions." These recommend; we ship a metric framework with reference implementation of those recommendations on a shared station.

B Per-Cell RMST and Pairwise P-P Plots

This appendix supports two claims from the body text. First, the per-(model, object) RMST grid (Figure 4) substantiates §5.1's 19% per-object HRT ceiling and surfaces the GR00T-vs-ACT object crossing referenced in §5.3. Second, the pairwise P-P plots without the human anchor (Figure 5) accompany §5.3's discussion of how the human-anchored AUC saturates and what removing that anchor reveals about the top-3 pairs.

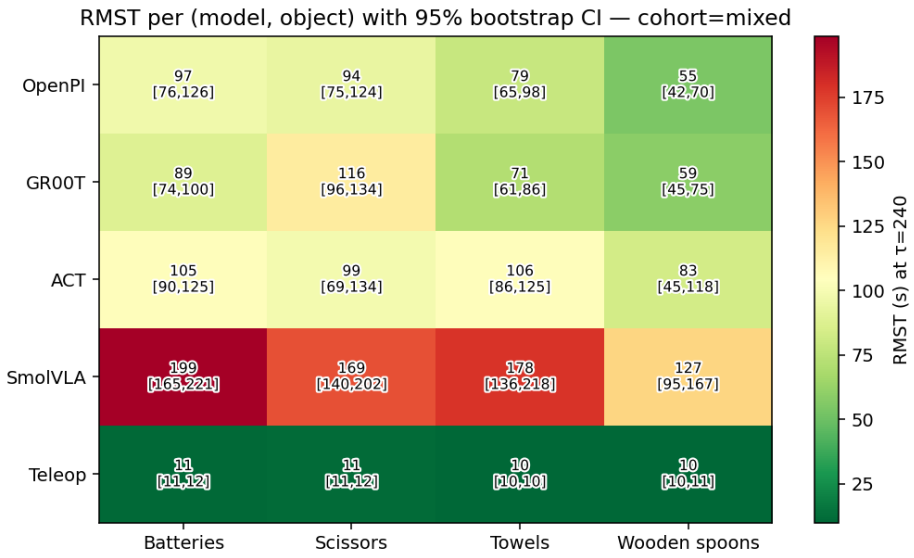


Figure 4: Per-(model, object) RMST with 95% episode-clustered bootstrap CIs. Crossings exist (most clearly GR00T vs. ACT between Batteries and Scissors) but none clears Bonferroni at $\alpha = 0.05/24$. The visible patterns suggest the rankings vary by object; resolving them formally requires the rollout budget that Figure 3 quantifies.

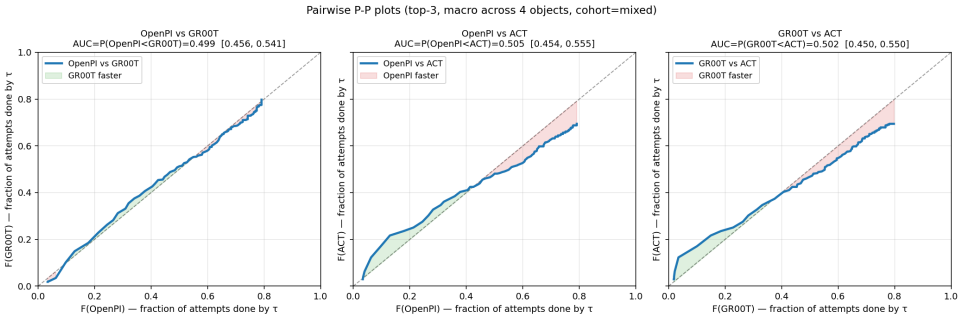


Figure 5: Pairwise model-vs-model P-P plots, top-3 only (SmolVLA excluded as too easily separable to be informative), macro-averaged across the 4 trained objects. The teleop-anchored AUC of §5.3 saturates because the human reference is $\sim 7\times$ faster than any evaluated VLA – nearly all model events fall in the human's tail, so the AUC integral is dominated by short-time behaviour. Removing the human anchor neutralizes that bias: every top-3 pair falls on or near the diagonal, with macro-AUCs all consistent with 0.5 (ACT-vs-OpenPI 0.51 [0.45, 0.56]; ACT-vs-GR00T 0.50 [0.45, 0.55]; OpenPI-vs-GR00T 0.50 [0.46, 0.54]).

C Run-Explorer Visualization

Every evaluated rollout is published in a run-explorer interface (Figure 6) that synchronizes the exterior and wrist video streams against the 3D end-effector trajectory and the per-channel telemetry: commanded action, gripper target, and measured gripper position. A reader auditing a specific time-to-success annotation against the raw video can replay it frame-accurately and inspect the joint state at any instant. The interface is built on Rerun [8] and runs post-hoc against the released artifacts; it is independent of the operator-side tooling used during rollout collection. We expose this not as part of the methodological contribution but as a transparency mechanism: every annotation in the released dataset is auditable against the raw rollout it was derived from.

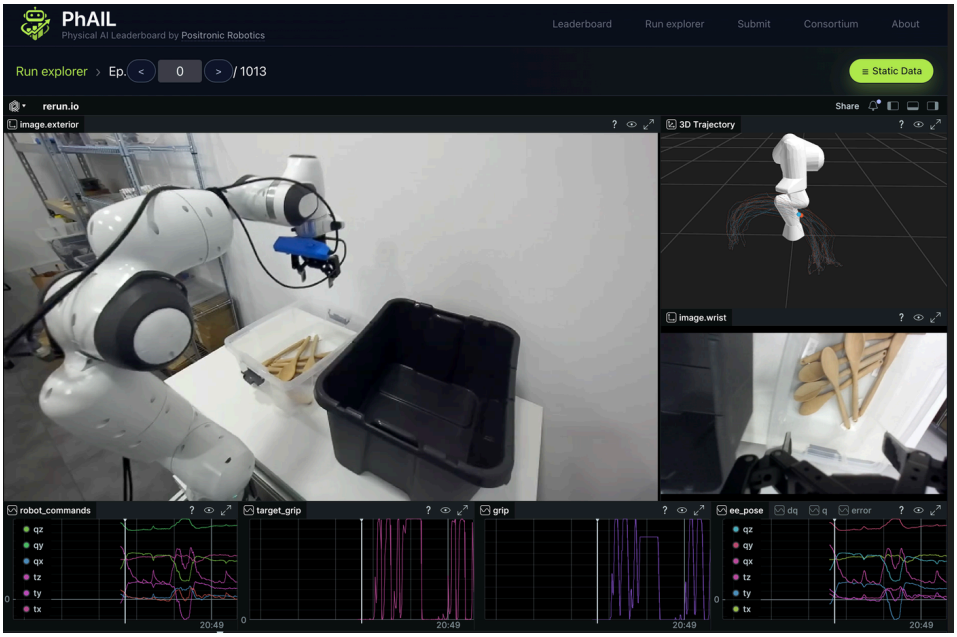


Figure 6: PhAIL platform during a rollout, visualized in the run-explorer interface (built on Rerun [8]): synchronized exterior and wrist video, 3D end-effector trajectory, and telemetry traces (commanded action, gripper target, measured gripper).

D Fine-Tuning Dataset Composition

Table 6 summarizes the shared fine-tuning corpus used across all four VLAs (§4.2): 352 demonstration episodes, ∼13 hours of robot operation, collected on the same Franka FR3 / Robotiq 2F-85 / dual-camera station as the evaluation rollouts. Episode counts are imbalanced across the four trained objects (wooden spoons are over-represented as the early development task); the full dataset is released alongside the benchmark.

Object	Episodes	Duration (min)
Wooden spoons	167	∼340
Towels	112	∼178
Scissors	83	∼160
Batteries	88	∼134
Total	352	∼812

Table 6: Fine-tuning dataset (released alongside the benchmark, ∼13 hours of robot operation).

E Spatial Configuration Sensitivity

PhAIL's protocol logs the spatial configuration (camera and tote placement) with every episode. Between rollouts we vary outbound tote placement (left/right) and external camera position (left/right), creating four spatial configurations. Table 7 reports completion rate by camera/tote configuration. "Same-side" means the external camera and the outbound tote are on the same side of the workspace; "opposite-side" means they are on opposite sides, in which case the inbound tote partially occludes the outbound tote from the camera's perspective.

Model	Same-side	Opposite-side	Δ
OpenPI	51.1%	45.0%	+6.1 pp
GR00T	57.8%	35.6%	+22.2 pp
ACT	35.6%	31.6%	+4.0 pp
SmolVLA	12.1%	9.5%	+2.6 pp

Table 7: Completion rate by camera/tote configuration.

GR00T shows the largest sensitivity (22.2 pp drop), suggesting strong reliance on the external camera view; OpenPI is the most robust among the top models. An environmental change a human operator would not notice can move the measured ranking. PhAIL's protocol logs the configuration with every episode so that this kind of confound is visible to the analyst rather than absorbed into noise.

F Power-Calculation Derivations

Wilson confidence interval on a single binary success rate. For an observed proportion $\hat{p}$ on N trials, the Wilson [5] score interval gives a CI of half-width approximately $z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/N}$ for the underlying success probability. To obtain ± 5 pp at 95% confidence on $\hat{p} = 0.70$, the formula gives $N \approx 380$ rollouts. At the field-modal $N \in [10, 20]$, the Wilson 95% CI on an observed 70% success rate is roughly $[0.40, 0.89]$ at N=10 or $[0.48, 0.86]$ at N=20 – intervals so wide that ranking claims at this depth are not statistically defensible.

McNemar test for paired binary outcomes. For two policies evaluated on the same N paired conditions, let b and c be the discordant cell counts (one policy succeeds, the other fails). The McNemar [6] statistic is $(b-c)^2/(b+c)$, χ^2_1-distributed under H_0 of equal success probabilities. Following Connor's [7] sample-size derivation for the paired-sample design, to detect a $\Delta = 5$ pp paired difference at 80% power and $\alpha = 0.05$, with discordance rate $p_d \in [0.10, 0.25]$ (capturing "how often the two policies disagree on a given paired condition"),

$$N \approx \frac{\left(z_{\alpha/2}\sqrt{p_d} + z_{\beta}\sqrt{p_d - \Delta^2}\right)^2}{\Delta^2}.$$

At $p_d = 0.10$ this gives $N \approx 600$; at $p_d = 0.25$, $N \approx 1500$. The range 600–1500 cited in the body reflects the realistic span of p_d for top-3 frontier-VLA pairs on our data; a discordance rate of 0.20 gives the central $N \approx 1280$.

Empirical detection-rate-vs-N curves. §5.2 translates these analytical numbers into empirical detection-rate-vs-N curves on our data via 300 outer subsampling trials, each running a 200-rep inner episode-clustered bootstrap. The full $(N, \text{metric}, \text{pair}, \text{detection rate})$ table for all four metrics ($F(30\text{s})$, $F(60\text{s})$, RMST, KS) and all six model pairs is released alongside the data.

G Annotation Protocol

The framework requires per-episode event logs: place timestamps, intervention counts, and end-state item counts. The protocol is implementation, not contribution, and is expected to evolve.

Automated detector + classifier. A multistage release detector reads gripper telemetry (`target_grip` edges gated by hold and plateau windows, plus accidental-drop detection on `grip` slumps from the held band into the fully-closed band) and emits per-candidate features (end-effector zone, hold duration, displacement, a running target-zone ledger); a classifier then assigns each candidate a verdict $\in$ {success, failure, neutral, unknown}. Implementation, calibrated thresholds, and unit tests are released alongside the data.

Manual review. For every episode where the classifier's success count disagrees with the operator's logged `eval.successful_items`, a human annotator reviews each candidate in a custom desktop UI and records per-candidate item counts and timestamps; these confirmed values ground

the released annotations. Subsequent PhAIL releases will replace this step with task-specific hardware sensing (e.g., bin-weight sensors for pick-and-place, analogous instrumentation for insertion and assembly) so that operation events are recorded objectively, with no manual-review bias.

H Reproducibility

Software environment: Python 3.13, Linux. Random seeds: 0 throughout. Expected runtimes on a 12-core workstation: figure regeneration $\sim$10 minutes; sample-efficiency experiment (300 outer $\times$ 200 inner bootstrap) $\sim$7 minutes. The full evaluation framework, fine-tuning recipes, dataset, per-rollout artifacts, and figure-regeneration scripts are released at URL withheld for double-blind review.

NeurIPS Paper Checklist

1. **Claims.** *Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?*
   Answer: Yes. The four numbered contributions in §1 (canonical primitive, aggregation-rule disagreement, sample-efficiency from the right test, open benchmark) each map to a specific results subsection (§5.1, §5.3, §5.2, §4.1).
2. **Limitations.** *Does the paper discuss the limitations of the work performed by the authors?*
   Answer: Yes. §6 discusses single embodiment, single task primitive (pick-and-place), $N \approx 35$ depth limits, and the constraint that all rankings reflect each model's default open-source fine-tuning recipe rather than per-model optimization.
3. **Theory, Assumptions, and Proofs.** *For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?*
   Answer: N/A. The paper is empirical and methodological. The statistical estimators used (Kaplan-Meier, restricted-mean survival time, Kolmogorov-Smirnov, episode-clustered bootstrap) are standard and cited (§3.1, §3.3); we do not introduce new theoretical results.
4. **Experimental Result Reproducibility.** *Does the paper fully disclose all the information needed to reproduce the main experimental results?*
   Answer: Yes. §4 documents the protocol, models, fine-tuning recipes, and cohort. §3 formalizes the methodology (estimator, ghost-event handling, bootstrap), Appendix F the power-calculation derivations, Appendix G the annotation pipeline, and Appendix H the software environment, seeds, and runtimes.
5. **Open Access to Data and Code.** *Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?*
   Answer: Yes. The benchmark, evaluation framework, dataset, and per-rollout artifacts (synchronized video, telemetry, annotations) are released at URL withheld for double-blind review. The reference implementation is open source.
6. **Experimental Setting/Details.** *Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?*
   Answer: Yes. §4.2 specifies that each of the four models was fine-tuned with its repository's default published recipe; we did not perform per-model hyperparameter search. Appendix D documents the fine-tuning dataset composition.
7. **Experiment Statistical Significance.** *Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?*
   Answer: Yes. All reporting scalars and CDF curves carry 95% paired episode-clustered bootstrap intervals; pairwise tests use stratified logrank with Bonferroni correction (§5.1, Table 3); the sample-efficiency study reports detection-rate-vs-N curves with 300 outer trials and 200-rep inner episode-clustered bootstrap (§5.2).
8. **Experiments Compute Resources.** *For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?*
   Answer: Yes. Inference for each VLA was run on the evaluation workstation that hosts

the Franka FR3 controller; statistical analyses run in $\sim$10 minutes (figure regeneration) and $\sim$7 minutes (sample-efficiency experiment) on a laptop (Appendix H). No GPU cluster is required to reproduce the analysis pipeline.

9. **Code of Ethics.** *Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?*
   Answer: Yes. The work is benchmark methodology on a controlled laboratory robot evaluating publicly released model checkpoints. No personally identifiable information is collected; no human subjects beyond paid task annotators reading robot video.

10. **Broader Impacts.** *Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?*
    Answer: Yes. The paper's intended societal impact is improved evaluation rigor for real-robot VLA policies, enabling more honest comparisons in the field. Potential negative impact is limited because no new model checkpoints are released; the contribution is methodological. The benchmark could in principle be used to falsely advertise weak models with cherry-picked metrics, which is exactly the practice the methodology is designed to surface.

11. **Safeguards.** *Does the paper describe safeguards that have been put in place for responsible release of data or models with a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?*
    Answer: N/A. No new model is released. The dataset consists of robot rollout videos and telemetry on a controlled bin-to-bin pick-and-place task with no high-risk content.

12. **Licenses for Existing Assets.** *Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?*
    Answer: Yes. All four evaluated VLAs (OpenPI $\pi_{0.5}$, GR00T N1.6, ACT, SmolVLA) are publicly released under their respective open-source licenses and cited (§4.2); the DROID-style hardware configuration is cited; the Rerun visualizer is cited.

13. **New Assets.** *Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?*
    Answer: Yes. The PhAIL benchmark is released with per-rollout video, telemetry, dataset documentation (Appendix D), annotation protocol (Appendix G), and a Croissant metadata file. License is open and declared in the dataset card.

14. **Crowdsourcing and Research with Human Subjects.** *For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?*
    Answer: Yes. Appendix G describes the Toloka high-tier crowd-annotation task, the three intervention classes annotators were trained on, and the curated 10-episode pilot used for instructions. Annotator compensation followed Toloka's high-tier rate schedule.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects.** *Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether IRB approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?*
    Answer: N/A. The crowd-annotation task involves paid microtask workers viewing pre-recorded robot videos; no protected-class data, biometric data, or personally identifiable information is collected. The activity does not constitute human-subjects research under standard IRB definitions.

16. **Declaration of LLM Usage.** *Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?*
    Answer: N/A. The evaluation methodology and statistical analyses do not involve LLMs. Some of the evaluated VLA architectures use language-conditioned components internally, but those are properties of the evaluated artefacts, not of our methodology.